INCH-POUND
MIL-P-20632B(SH)
20 April 1992
SUPERSEDING
MIL-P-20632A(SHIPS)
13 May 1955
(See 6.14)

MILITARY SPECIFICATION

PURIFIERS, CENTRIFUGAL, FUEL AND LUBE OIL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fuel and lube oil centrifugal purifiers for service on board naval ships.

1,2 Classification. Centrifugal purifiers are of the standard sizes specified in table I (see 6.2).

TABLE I. Purifier size.

Capacity 1/			
Standard size	Turbine lube oil MIL-L-17331, 2190-TEP at 71°C (160°F)	Naval distillate fuel F-76, MIL-F-16884 at 21°C (70°F)	Diesel lube oil MIL-L-9000 at 79°C (175°F)
B	852 (225)	946 (250)	284 (75)
C	1,324 (350)	1,608 (425)	473 (125)

1/ Capacity in liters per hour (gallons per hour),

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter

AMSC N/A FSC 4330
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting.
QQ-S- 766	- Steel, Stainless and Heat Resisting, Alloys, Plate, Sheet and Strip
PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

MILITARY

MIL-S-901	- Shock Tests, H.I (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-E-917	- Electrical Power Equipment, Basic Requirements (Naval Shipboard Use)
MIL-C-2212	- Contractors and Controllers, Electric Motor AC or DC, and Associated Switching Devices
MIL-G-2860	- Glasses, Sight-Flow, Clear, Borosilicate
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 JP-5 and JP-5/JP-8 ST
MIL-L-9000	- Lubricants Oil , Shipboard Internal Combustion Engine, High Output Diesel
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment
MIL-P-15024/5	- Plates, Identification
MIL-P-16789	- Pumps (Including Prime Movers and Support Items), Packaging of
MIL-F-16884	- Fuel, Naval Distillate
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral - Horsepower, Shipboard Use
MIL-I-17244	- Indicators, Temperature, Direct -Reading. Bimetallic, (3 and 5 Inch Dial).
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear , Moderate Service
MIL-M-17413	- Motors, Direct Current, Integral H P., Naval Shipboard
MIL-H-17428	- Heaters, Fluid, Industrial (Lubricating Oil) (for Naval Shipboard Use).
MIL-M-17508	- Mounts, Resilient Types 6E100, 6E150, 7EL50, 6E900, 6E2000, 5E3500, 6E100BB, 6E150BB, 7E450BB, and 6E900BB
MIL-B 17931	- Bearings, Ball, Annular, for Quiet Operation

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-G-18997 - Gauge, Pressure, Dial Indicating.
MIL-P-19131 - Pumps, Rotary, Power Driven, Miscellaneous .
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-S-24093 - Steel Forgings, Carbon and Alloy Heat Treated.
MIL-H-24299 - Heaters, Lubricating Oil, Electric, Naval Shipboard.
MIL-P-24441 - Paint, Epoxy-Polyamide General Specification for.
MIL-P-24441/1 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I,
MIL-P-24441/2 - Paint, Epoxy-Polyamide, Exterior Topcoat, Haze Gray, Formula 151, Type I.
MIL-P-24691 - Pipe and Tube, Carbon, Alloy and Stainless Steel, Seamless and Welded, General Specification for.
MIL-P-24691/3 - Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded
MIL-C-24707 - Castings, Ferrous, General Specification for.
MIL-C-24707/1 - Castings, Ferrous, for Machinery and Structural Applications.
MIL-C-24707/3 - Castings, Ferrous, Corrosion-Resistant, Austenitic, Chromium-Nickel .
MIL-C-24707/5 - Castings, Ductile Iron and Austenitic Ductile Iron
MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and Facilities.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-H28/2 - Screw-Thread Standards for Federal Services Section 2 Unified Inch Screw Threads - UN and UNR Thread Forms

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarines Service
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470 - Maintainability Program for Systems and Equipment.
MIL-STD-721 - Definitions of Terms for Reliability and Maintainability.
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

MILITARY (Continued)

MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.

HANDBOOK

MILITARY

MIL-HDBK-267 - Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications The following other Government documents , drawings, and publications form a part of this document to the extent specified herein Unless otherwise specified, the issues are those cited in the solicitation

DRAWINGS

MILITARY
NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 803-1385850 - Piping Instrument Pressure for All Services
NAVSHIPS 810-1385917 - Temperature Indicator and Thermowell Selection

(Application for copies should be addressed to Commander, Portsmouth Naval Shipyard, Code 202 2, Portsmouth, NH 03801)

2. 2 Non-Government publications The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 29 - Standard Specification for Steel Bars, Carbon and Alloy, Hot-Wrought and Cold-Finished, General Requirements for (DoD adopted)
A 108 - Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality. (DoD adopted)
B 26 - Aluminum Alloy Sand Castings
D 2276 - Standard Test Method for Particulate Contaminants in Aviation Turbine Fuels. (DoD adopted)
D 2709 - Standard Test Method for Water and Sediment in Distillate Fuels by Centrifuge. (DoD adopted)
F 1166 - Human Engineering for Marine Systems Equipment, and Facilities

Source: https://assist.dla.mil -- Downloaded: 20261O-19T08:08Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)
SP 10 - Near-White Blast Cleaning. (DoD adopted)

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Materials. Purifier materials shall be as specified in table II. However, this specification is not intended to be restrictive, provided proposed alternate materials are equal to or better than the materials specified. Proposed alternate materials shall be subject to review and acceptance by the contracting activity. Parts for which specific materials are not specified shall be made of materials similar to those cited for the service intended. Use of dissimilar materials shall be avoided.

TABLE II. Purifier materials.

Application	Material	Specification
Frames and covers	Stainless steel Cast aluminum	QQ-S-766, class 304 ASTM B26
Bowl assembly	Stainless steel Stainless steel casting	QQ-S-763, class 304, 316, 410 QQ-S-766, class 316 MIL-C-24707 and MIL-C-24707/3
Spindle	Forged steel, free machining Cold rolled steel	MIL-S-24093, class B ASTM A 108

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

TABLE II. Purifier materials - Continued.

Application	Material	Specification
Spindle housing	Nodular iron Cast steel	MIL-C-24707/5 MIL-C-24707/1, A 757, Gr A1Q or A 216, Gr WCA
Inlet and outlet piping	Stainless steel	MIL-P-24691 and MIL-P-24691/3, grade TP304

3 2 1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

3 2 2 Prohibited materials. The following materials are prohibited for use in purifiers covered by this specification

(a) Magnesium and its alloys
(b) Cast iron (except nodular iron)
(c) Semi-steel.
(d) Cadmium plating
(e) Copper alloys in contact with fuel or lubricating oil
(f) Asbestos

3 2.3 Mercury. The purifier shall be free of mercury contamination. During the manufacturing process, tests and examination, the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment

3. 3 Construction Purifiers shall separate contaminants f-rem F.76 and JP-5 fuels, turbine lube oil, and diesel engine lubricating, oil. Purified oil or fuel shall be discharged under pressure through closed piping. to the ship's fuel or lube oil systems, as applicable Water separated from the oil or fuel shall be continuously discharged to ship's oily water drain collection system. Separated solids shall collect within the bowl until they are removed by manual cleaning

3.3 1 General Purifiers shall have the following characteristics

(a) Manual pushbutton motor start
(b) Manual control for flow of oil after prime
(c) Electric motor drive through belts or gears
(d) Attached feed pump, discharge pump, and piping arrangement as indicated on figure 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

3.3.2 Mounts. Purifiers shall be vertical and hard-mounted on a base plate and furnished complete, ready for mounting directly to the deck of the ship without the use of a pedestal, unless the use of resilient mounts is cited in the specific acquisition (see 3.3.2.1).

3.3.2.1 Resilient mounts. When specified in the contract or order (see 6.2), resilient mounts shall be provided in accordance with MIL-M-17508.

3.3.3 Frames. Purifier frames shall be sufficiently rigid to permit lifting, handling, shipment, and installation of the units on board ship without disturbing the alignment of the assembled units, and so that the normal distortion, weaving, or vibration of the supporting stricture on board ships will not cause misalignment of rotating elements.

3.3.3.1 Special handling provisions. Special tools or fittings shall be provided by the manufacturer for lifting all items weighing in excess of 16 kilograms (35 pounds).

3.3 4 Brakes. Purifiers shall be equipped with bowl brakes.

3.3.5 Bearings. Purifier bowl spindle bearings shall be ball bearings. Other bearings may be of the sliding contact or rolling contact types. Double row bearings and bearing sets are prohibited unless they are the self-aligning type.

3.3.5.1 Ball bearings, Ball bearing installation and lubrication shall be in accordance with FF-B-171 for surface ships. For submarines and low noise surface ship applications, bearings shall be in accordance with MIL-B-17931, except bearings operating at speeds greater than 3600 revolutions per minute (r/min) need only meet the vibration requirements of MIL-B-17931. Bearing assemblies shall meet the vibration requirements of MIL-B-17931.

3.3.6 Lubricants. Lubricants used for purifier operation, assembly, and maintenance, except high speed (above 3600 r/rein) sealed bearings, shall be selected from those standard lubricants listed in MIL-HDBK-267. Commercial lubricants may be used for sealed bearings operating above 3600 r/min.

3.3.7 Piping and valves. Connections to ship’s piping shall be flanged. Flange size, rating, specification, and surface finish for piping connections shall be as specified (see 6.2). Purifier valves, flanges, bolting, gaskets, fittings, and piping shall conform to MIL-STD-777 for surface ship applications and MIL-STD-438 for submarine applications (see 6.2).

3.3.8 Prime water connections. A connection shall be provided on purifiers for purifier bowl prime. Prime water will be provided from the ships potable water system.

3.3 9 Drain connections. Connections shall be provided for casing or funnel drains as applicable,

3.3.10 Flow adjustment. Purifiers shall be equipped with a means of adjusting the oil or fuel flow rate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

3.3.11 Observation ports and sight glasses Observation ports and sight glasses shall be provided in the clean oil or fuel discharge and the water drain line Sight glasses shall be in accordance with MIL-G-2860

3 3,12 Pressure gauges, gauge piping, and connections Pressure gauges shall be 9 centimeters (cm) (3.5 inches) in diameter and shall be in accordance with MIL-G-18997 Gauge piping and connections shall be in accordance with Drawing 803-1385850

3 3.13 Thermometers. Thermometers shall be 7 6 cm (3 inches) in diameter, in accordance with MIL-T-17244. Use of mercury thermometers is prohibited.

3 3.13.1 Thermometer wells Thermometer wells and bulbs, when supplied, shall be in accordance with Drawing 810-1385917.

3 3.14 Oil dipstick and sight glass. Gear driven purifiers shall be provided with an oil dipstick or sight glass to determine the lube oil level within the drive mechanism case

3.3.15 Thread fit. Screw threads shall be in accordance with FED-STD-H28 and FED-sTD-H28/2

3 3.16 Doweling Each component part of an assembled unit supported directly by a frame or bedplate shall be doweled to facilitate reassembly and maintenance of alignment Dowels shall be unequally spaced or be of different diameters, to avoid incorrect assembly.

3.3.17 Machined surfaces. Bearing and seating surfaces of frames or bedplates shall be finish machined.

3.3.18 Welding. Welding shall be in accordance with MIL-STD-278 Loose welding beads, slag, and spatter shall be removed.

3.3.19 Painting. Non-working surfaces, normally painted, shall be finished as follows.

(a) Surface preparation prior to painting in accordance with SSPC SP 10
(b) Two coats epoxy paint formula 151 and one coat of formula 150 in accordance with MIL-P-24441, MIL-P-24441/1, and MIL-P-24441/2 Average 3 mils dry film thickness per coat

3.3.20 Purifier pumps Purifier pumps shall be type III, class N-4 or 0-3 in accordance with MIL-P-19131 The capacity of each discharge pump shall be approximately 110 percent of the rated capacity of the corresponding feed pump Pumps shall be furnished with built-in spring loaded bypass valves A pressure gauge and thermometer shall be installed between the discharge of the feed pump and the inlet to che purifier bowl The suction lift of the pumps, referred to the horizontal centerline of pump rotors, shall be 15 inches of mercury The discharge pressure of pumps shall be 25 pounds per square inch (lb/in^2) gauge when operating under the specified conditions of service.

Source: https://assist.dla.mil -- Downloaded: 2026t0-19T08:08Z
Check the source to verify that this is the currentversion before use.

3.3.20.1 Purifier pump drive Unless otherwise specified (see 6.2), the pumps and the purifier shall be driven by a common motor.

3.3.21 Oil heaters. When specified in the contract or order (see 6.2), oil heaters shall be supplied with the purifiers. Oil heaters shall heat oil from the temperature specified in the contract or purchase order (see 6 2) , to the purification temperature specified in 3.5 1, with the oil flowing through the heater at the rated capacity of the feed pump

3.3.21.1 Oil heater type. Oil heaters shall be in accordance with MIL-H-17428 (steam type) or MIL-H-24299, type B (electric type) (see 6.2).

3.3.22 Motors. Motors shall conform to MIL-M-17060 for alternating current (at) motors and shall be of the following characteristics:

Horsepower (hp)	See 3.3.22.2.
Service	A.
Speed classification	Constant.
Duty	Continuous
Enclosure	Totally enclosed or dripproof.
Bearings	In accordance with FF-B-171; MIL-B-17931 if low noise application
Ambient temperature	50 degrees Celsius (°C) (122 degrees Fahrenheit (“F)).
Insulation	B or F with sealed insulation system
Type	Squirrel cage induction.
Voltage rating	440 volts ac, 3-phase, unless otherwise specified in the contract or order (see 6.2).
Design	As required (see 3.3.24)
Low noise application	As required in the contract or order (see 6.2).
Mechanical balance	Precision; superprecision if low noise application

3.3.22.1 Direct current (dc) motors. When specified in the contract or order (see 6.2), dc motors shall be provided in accordance with MIL-M-17413.

3.3.22.2 Horsepower (hp) rating. The hp rating of each motor shall be not less than the sum of the maximum brake hp of the driven components under any condition of service, The actual motor rating shall be in accordance with Navy standard hp sizes in accordance with MIL-M-17060 or .MIL-M-17413, as required. If the maximum brake hp of the equipment is less than 2, and if there is a possibility of an unpredictably high frictional load due to improper adjustment of glands or to some other cause, then the maximum brake hp shall be multiplied by a safety factor of [1 5 - (Max. BHP)/4] before selecting the next larger Navy standard rating

3.3.23 Controllers. When specified in the contract or order (see 6 2), motor controllers shall be provided. Motor controllers shall conform to MIL-C-2212 and shall be of the following characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

Enclosure	Dripproof
Operation	Magnetic.
Type	Acrossline for ac - Resistor for dc
Function	Motor starting,
Duty	Continuous.
Protection	Overload - under voltage (LVP).
Ambient temperature	50°C (122°F).
Voltage	440 volts ac, 3-phase, unless otherwise indicated (see 6.2) (dc voltage requirements, if applicable, shall be as specified in the contract or order (see 6 2)).
Performance	Manual
Master switches	Local, manual.
Abnormal vibration frequencies	When not covered by MIL-C-2212, frequency will be indicated in the contract or order (see 6 2).
Low noise application	When required in the contract or order (See 6 2), insulation shall be class B with class A for coils only.

3.3.24 Acceleration allowance. Motor and controller shall permit acceleration of the unloaded purifier to full operating speed in less than 3 minutes without exceeding permissible temperature rise in any components or causing wear due to slippage of drive

3.3.25 Accessibility. The construction shall provide the maximum accessibility (within space limitations) to parts which require scheduled maintenance and to effect repairs. Construction shall be so that the purifier car-i be opened for cleaning and maintenance without breaking any piping connections Removal of major operational parts shall not be necessary to permit accessibility for maintenance, inspection, and repair. The human engineering, considerations specified in MIL-STD-1472 and MIL-H-46855 shall not be degraded during detail design and construction.

3.3.26 Interchangeability, Identically identified components shall be functionally and physically interchangeable, without degradation of performance, reliability, or operating characteristics and without selective assembly or modification except for calibration and adjustment Repair parts shall be interchangeable with, and identified identically to, the part they replace Notwithstanding the above, parts balanced as complete assembles shall not be exchanged without rebalancing the complete assembly.

3.4 Performance

3.4.1 Rated capacity The capacity of a purifier shall be the flowrate stated in table I, maintained for a period of 1 hour, during which time the purified fluid shall show not more than O 05 percent water by volume nor more than 0.02 percent solids by volume (maximum particle size 5 micrometers.)

3.4 2 Emergency capacity The purifier shall efficiently remove the water from a mixture that is half oil or fuel and half water. The purified fluid shall contain not more than 0.3 percent water by volume at 50 percent rated capacity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

3 4.3 Solids retention. The solids retention volume of the purifier shall be not less than stated in table III.

TABLE III. Solids retention capacity.

Size	Volume (liters (quarts))
B	0.87 (0.92)
C	1.23 (1.30)

3.4.4 Shock. The purifier shall meet the requirements for grade A, type A, class I or II (see 6.2) high impact shock in accordance with MIL-S-901. The purifier shall restart and resume normal operation if it shuts down when subjected to the test requirements of MIL-S-901. Equipment shall be hard-mounted to a suitable base plate for test purposes. The determination of stresses resulting from the shock test shall be accomplished by the application of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (O 2 percent offset of the material) Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses The maximum accelerations possible under the shock tests shall be withstood by hold-down bolts, supporting feet or lugs, and main structural member of the equipment. No failure or permanent deformation may occur as a result of shock. Components of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless there is reason to believe that higher accelerations will be experienced by the components.

3 4,5 Mechanical vibration. The equipment shall conform to the requirements of MIL-STD-167-1, type II.

3.4.5.1 Environmental vibration The equipment shall withstand the environmental vibration requirements of MIL-STD-167-1, type I

3.4.6 Noise. The equipment shall be in accordance with the acceptance criteria specified in MIL-STD-740-1 (airborne) and MIL-STD-740-2 (structureborne), when tested (see 4 6.12)

3.4.7 Electromagnetic Interference. The equipment shall be in accordance with electromagnetic emission suppression requirements of MIL-STD-461, parts 1 and 5, class A4, for surface ships Acceptance testing shall be in accordance with MIL-STD-462

3 5 Environmental conditions. Equipment shall operate continuously at rated capacity under any combination of the range of environmental conditions specified herein

3.5.1 Temperatures. Table IV establishes the applicable temperature ranges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

TABLE IV Temperature 1/

	Naval distillate, F-76	Turbine lube oil	Diesel lube oil
Influent temperature range	16-27 (60-80)	66-77 (150-170)	74-85 (165-185)
Ambient temperature range	4-50 (40-122)	4-50 (40-122)	4-50 (40-122)

1/ Temperatures expressed as °C (°F)

3.5.2 Pitch and inclined operation. The equipment shall suffer no damage and shall operate to the requirements of the specification when subjected to the motions stated in table V

TABLE V Motions 4/ 5/

Motion attitude	Surface ships (radians (degrees))	Submarines (radians (degrees))
Pitch (from normal either direction) 1/	0 17 (10)	0 17 (10)
Trimmed down (from normal horizontal)	0 087 (05)	0 52 (30)
Roll (from vertical) 2/	3/ 0 79 (45)	1 05 (60) surfaced 0 52 (30) submerged
Permanently listed (either side of vertical)	0 26 (15)	0 26 (15)

1 Time of cycle: 10 seconds
2/ Time of cycle: 15 seconds
3/ The purifier shall maintain lubrication without loss of oil to the drive assembly The purifier shall continue to operate without the need for re-priming after recovering from the subject motion attitude
4/ Trim and list conditions shall be considered to occur simultaneously
5/ Roll and pitch conditions shall be considered to occur simultaneously

3.6 Reliability and maintainability Definitions of terms for reliability and maintainability shall be in accordance with MIL-STD-721, MIL-STD-785, and MIL-STD-470.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

3.6.1 Reliability Mean time between failure (MTBF) for the purifier shall be not less than 1500 hours (see 6.3)

3 6 1.1 Failure. For purposes of reliability calculation, the purifier is in a failed state if it cannot be started, or if a malfunction occurs that forces the purifier to be shut down prior to the completion of its scheduled test time or operating time. In addition, the purifier shall be considered to fail if the capacity (see 3,4.1) or the quality of purification (see 4.6 8) is not in accordance with the requirements specified, even if the unit has not experienced a forced shutdown. Failures shall be classified as either relevant or nonrelevant. Only relevant failures shall be chargeable for verifying reliability requirements.

(a) Relevant failures are caused by:

(1) Equipment design defects.
(2) Equipment manufacturing defects,
(3) Parts defects
(4) Unknown
(5) All other causes not specifically listed as non-relevant

(b) Non-relevant failures are caused by:

(1) Accident, mishandling, or improper storage
(2) Operator or procedural error
(3) External test equipment or- facility failures
(4) Drawing, maintenance manual, or other documentation errors, provided prompt correction is shown to eliminate future similar failures (this shall not be construed *to* include design errors),
(5) Failures of multiple, simultaneous or immediately sequential nature (only the initial failure shall be counted in such cases)
(6) Direct result of relevant or non-relevant failures of another item of equipment (secondary failures)
(7) Improper maintenance induced failures
(8) Foreign object damage, unless caused by failure of protective device built into seller’s equipment.

3.6.1.2 Failure mode, effects and criticality analysis (FMECA). A FMECA shall be performed (see 6 3 and 6 10)

3.6.2 Maintainability. Mean time to repair (MTTR) for the purifier shall not exceed 10 hours (see 6.3)

3.7 Identification plates. Identification plates shall be furnished on each purifier and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that plastic Identification plates are not acceptable Identification plates shall be secured to equipment with corrosion-reslscan~ metallic screws.

3.7.1 Identification information. Purifier identification plates shall contain the following information.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:08Z
Check the source to verify that this is the current version before use.

(a) Manufacturer's name.
(b) "Centrifugal Purifier" (followed by flow rate).
(c) Manufacturer's model.
(d) Navy size designation.
(e) Date of manufacture (month and year).
(f) Manufacturer's serial number.
(g) Contract number (and item number for multiple unit orders).
(h) National stock number.
(i) Nominal bowl speed.

3.7.2 Other identification plates. Each motor and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.8 Safety criteria. Equipment shall be constructed to ensure safety to operating and maintenance personnel. Purifiers shall have their bowls enclosed to prevent free vapor from escaping. External moving parts, which are a potential hazard to personnel, shall be avoided wherever practicable. When their use is unavoidable , positive protection in the form of a guard shall be provided. Sharp corners and projections, which may cause injury to personnel or on which clothing may catch, shall be avoided. The following are some of the known safety criteria that apply to auxiliary machinery; others are contained in documents such as MIL-STD-454 and MIL-E-917.

(a) Exposed rotating or reciprocating components, such as couplings, linkages, belts, chains, and flywheels shall be enclosed with a safety guard to prevent accidental personnel contact with the moving part.
(b) When an equipment specification requires the presence of a hazardous material as a component of shipboard equipment, conspicuous indication on the equipment of the presence of such material, together with necessary warnings and instructions, is required.

3.9 Special tools. Special tools shall be furnished in the quantity specified (see 6.2 and 6.11). Special tools are defined as those items required for shipboard maintenance, servicing, and repair not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Navy Fleet Material Support Office, Mechanicsburg, PA 17055-0787 (FLEMATSUPPO)).

3.10 Workmanship. All parts of the purifier, before and after painting, shall be clean and free of sand, rust, dirt, pits, sprues, scale, and other harmful extraneous material which might detract from the intended operation, function, or appearance of the equipment. Fasteners shall be fully engaged and bolt heads and nuts shall be flat on washers or mating surface.

4, QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system **or** quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4),

4.3 First article inspection. First article inspection shall be performed on a sample purifier produced with equipment and procedures used in normal production.

4,3.1 Sampling for first article inspection. One sample unit of each size (see table I) to be supplied shall be subjected to first article inspection. The sample purifier shall be submitted for first article tests with pumps attached and driven by the same motor that drives the purifier.

4.3.2 Inspection routine. The sample unit shall be subjected to the examinations and tests specified in 4.5 through 4.6.2 and 4.6.4 through 4.6.14 (see 6.3).

4.3.3 Failure. Failure to pass the examination (see 4.5) or failures other than those allowed during testing (see 3.6.1.1) shall be cause for rejection.

4.4 Quality conformance inspection. Every purifier shall be examined and tested as specified in 4.5 through 4.6.2 and 4.6.12.2; sample purifiers (see 4.4,1) shall be tested as specified in 4 6.3 (see 6.3).

4.4.1 Sampling plan for driver test. A random sample of purifiers shall be selected from each lot in accordance with table VI for the test specified in 4.6.3 (see 6.9).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

TABLE VI Sampling. plan, acceptance and rejection

Lot size	Sample size	Accept	Reject 1/2/3/
2 - 15	2	0	1
16 - 25	3	0	1
26 - 90	5	0	1

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until rejection criteria is reached,
3/ Rejected lots may be screened and resubmitted for inspection and retest.

4.5 Examinations. Every purifier shall be examined to ascertain that the equipment is in accordance with all the requirements of this specification not involving tests and as specified in 4.5.1 through 4.5.3.

4.5.1 Fabrication. Welding shall be examined to ensure conformance with 3.3 18

4.5.2 Painting. Every purifier shall be examined after the application of each coat of paint to ensure conformance with the requirements specified in 3 3 1 9

4.5.3 Workmanship. Every purifier shall be examined to ensure conformance with requirements for workmanship specified in 3 10

4.6 Tests. Purifiers shall be tested as specified in 4 6. 1 through 4 6 14

4.6.1 Spin-test Prior to Installation in the purifier, the bowl assembly shall be spin-tesceel at .125 percent of the maximum designed operating, speed for d period of not less than 15 minutes

4.6.2 Operational test. Every completely assembled purifier shall be operated for a period of not less than 10 minutes at rated speed to validate smoothness of running,, satisfactory operation of controls , and proper assembly

4.6.3 Driver test. The purifier shall be operated at full capacity, as specified in table I, for a period of not less than 1 hour Quantitative measurements of purification are not required for this test If the purifier is supplied with a motor controller (see 3.3.23), the motor controller supplied with the purifier shall be used during this test During this test, the brake horsepower input to the assembled purifier shall be determined The type of fluid circulated through the purifier during the test shall be the fluid specified for the particular size and capacity (see 3.4.1)

4.6.4 Capacity test. The sample unit shall be tested for capacity to ensure conformance with 3. 4 1. The purifier shall be operated using turbine lube oil , diesel lube oil and naval distillate fuel F-76, each for 1 hour, as specified in table I, to determine the purifier capacity For each fluid an effluent quality

Source: https://assist.dla.mil -- Downloaded: 20261O-19T08:08Z
Check the source to verify that this is the current version before use.

test shall be performed as specified in 4.6.8. A one-minute surge load test shall be performed on the sample unit and for each fluid tested, as specified in 4.6.9. Purifiers constructed to purify F-76 fuel shall purify JP-5 in accordance with MIL-T-5624 as an alternate fuel.

4.6.5 **Reliability test**. The sample unit shall be operationally tested for a total of not less than 1800 hours to ensure conformance to 3.6 (see 6.3). Each operational period shall be not less than 7 hours. The motor controller used during this test shall be identical to the one to be supplied with the purifier. The operating fluids during the test shall be turbine lube oil in accordance with MIL-L-17331 (reliability test may be conducted concurrently with the tests specified in 4.6.7 through 4.6.10). An effluent quality test shall be performed on the sample unit at 10, 600, 1200, and 1800 hours of operation (see 4.6.8).

4.6.6 Maintainability test. The contractor shall demonstrate that the equipment meets the quantitative requirements specified in 3.6.2 (see 6.3).

4.6.7 Inclination test. The purifier shall be at an inclination of 0.262 radians (15 degrees) for 24 hours of the reliability test,

4 6,8 Effluent Quality test. Test conditions, acceptable performance, and methods of determining acceptable performance shall be as specified in 4 6.8.1 through 4.6.8.3.

4.6.8.1 Test conditions, Test conditions shall be as follows:

(a) Flow. As specified in 3.4.1 plus or minus 5 percent or in the contract.
(b) Water in influent. The amount of water shall be varied from 0.5 to 4.0 percent by volume.
(c) Solids. The amount of solids shall be varied from 0.1 to 0.15 percent by volume.
(d) Temperature. For fuel: 21 ± 5°C (70 ± 10°F). For turbine lube oil: 71 ± 5.5°C (160 ± 10°F). For diesel engine lube oil: 79 ± 5.5°C (175 ± 10°F)

4.6.8.2 Acceptable performance. Acceptable performance shall be as follows:

(a) Solids in effluent. Less than 0.02 percent by volume (maximum particle size 5 micrometers).
(b) Oil or fuel in water effluent. Less than 0.05 percent by volume.
(c) Free water in purified effluent. Less than 0.50 percent by volume.

4.6.8 3 Test methods. The influent and effluent samples shall be taken simultaneously . Alternate or modified test procedures (i.e. sampling and analysis) shall be subject *to* review and acceptance by the contracting activity.

4.6.8.3.1 Influent test methods. Influent quality test methods shall be performed to verify conformance to 4.6,8.1 (b) and (c) as follows:

(a) Sampling procedures. The sample cock shall be fully opened and 200 ml shall be drawn. The 200 ml shall be discarded Samples shall then be obtained as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

(1) Undissolved (free) water (BS&W). At least six influent samples shall be drawn directly into a 100 ml graduated centrifuge tube for free water determination

(2) Gravimetric analysis At least six influent samples shall be drawn into a 32 oz. bottle for gravimetric analysis

(b) Undissolved [free) water determination (BS&W) Undissolved water shall be determined in accordance with ASTM D 2709 (i e centrifuge , 100 mil capillary tip test tubes) Undissolved water shall be determined to nearest 10 p/m The water in oil shall be as specified in 4.6.8.1.(b)

(c) Gravimetric analysis (solids content). Gravimetric analysis for solid contaminant shall be in accordance with ASTM D 2276 Analysis shall be performed on entire sample The solids contaminate injection shall be as specified in 4 6 8 1 (c)

4 6 8.3 2 Effluent test methods. Effluent quality test methods shall be as follows:

(a) Sampling procedures The sample cock shall be fully opened and 200 ml shall be drawn The 200 ml shall be discarded Samples shall then be obtained as follows

(1) Undissolved (free) water and gravimetric analysis A sample of at least 800 ml shall then be collected for analysis, without closing the sample valve At least 500 ml shall be used for gravimetric analysis and at least 300 ml used for determination of undissolved water

(2) Water rejection (oil in water) At least one sample shall be collected in accordance with ASTM D 3370 The sample shall be prepared in accordance with ASTM D 3921

(b) Gravimetric analysis (solids removal) Gravimetric analysis for solid contaminant shall be in accordance with ASTM D 2276

(c) Undissolved (free) water Undissolved water shall be derermined in accordance with ASTM D 3240 Undissolved water shall be determined to the nearest 1 p/m

(d) Water rejection (oil in water). The oil in-water analysis shall be determined in accordance with ASTM D 3921

4.6.9 Surge load test. The sample unit shall proccess influent with water concentration of 50 percent when operating at 50 percent of rated capacity The purified effluent shall show not more than 0. 3 percent water by volume

4.6.10 Solids retention test. The ability of the unit to remove solids and water shall not decline as the solids retention capacity is approached A solids retention test shall be performed during which effluent samples are taken at regular intervals as the purifier is run from a clean bowl condition to the point when the calculated solids retention capacity is approached These samples shall be analyzed as specified in 4.6.8.3. The analyses shall verify that all the samples comply with the performance specified in 4.6.8.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

4.6.11 Vibration test. The sample unit shall be subjected to the type I and II environmental and mechanical vibration requirements of MIL-STD-167-1 The frequency range for type I vibration shall be as specified in the contract or order (see 6.2).

4.6.12 Noise test. The purifier shall be noise tested as follows.

4.6.12.1 Airborne noise test. The sample purifier shall be tested in accordance with MIL-STD-740-l and shall meet the performance requirements specified in 3.4.6,

4.6.12.2 Structureborne noise test. When specified in the contract or order (see 6.2), every completely assembled purifier shall be tested in accordance with MIL-STD-740-2 and shall meet the performance requirements specified in 3.4.6.

4 6.13 Shock tests. The sample unit shall be subjected to the high impact grade A, type A shock test specified in MIL-S-901. The test shall be class I or class II in accordance with MIL-S-901 as specified in the contract or order (see 6.2 and 6.11)

4,6. 13.1 Resubmission for shock testing Equipment previously accepted will not be resubmitted for shock test, except when evidence of low shock resistance develops in the units installed The contracting activity will initiate action with the contractor for correction of deficiencies, and may require shock tests at any time.

4.6.13 2 Shock testing with drivers Unless otherwise specified (see 6.2), the test purifiers shall be shock tested with drivers Purifiers shock tested with one driver will not be required to be shock tested again when supplied with a different shock qualified driver, Drivers are subject. to shock tests in accordance with the applicable equipment specifications.

4.6.13.3 Failure criteria. A purifier shall be considered *to* have failed the shock test in the event of any of the following

(a) Breakage of any part, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as a shaft, mounting feet, bearing (except for anti-friction bearing brinelling), or other part which would render the unit unusable in an emergency
(c) A mechanical imbalance of more than two times the amplitude of balance measured prior to test at rated speed
(d) The purifier cannot be restarted or does not exhibit acceptable performance as specified in 4 6 5

4.6.14 Electromagnetic Interference. The purifier shall be tested as specified in MIL-STD-462, to determine conformance to 3 4 7

4.7 Inspection of packaging, Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging The packaging (preservation, packing, and marking) requirements of purifiers shall be in accordance with MIL-P-16789 for the level (A, C, or commercial) of preservation; level of packing (A, B, C, or commercial) , and marking, including packaging acquisitioning options therein as specified (see 6,2). In addition, the following applies

(a) Navy fire-retardant requirements.

(1) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood, including laminated veneer material used in shipping container and pal let construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows

Levels A and B - Type II - Weather resistant
Category 1 - General use.
Level C - Type I - Non-weather resistant
Category 1 - General use

(2) Fiberboard Fiberboard used in the construction of class domestic. non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320

5.1.1 Marking. In addition to the marking required under 5.1, purifiers s'nail be marked as follows

(a) Critical close tolerance equipment. Unit packs and shipping containers and unpacked shipments shall be marked with the following

CRITICAL, CLOSE TOLERANCE OPERATING EQUIPMENT
HANDLE WITH CARE
DO NOT DROP OR SUBJECT TO SHOCKS OF JARS

Markings shall be stencil led, red color, and applied on two sides and both ends of the container or shipment. Letters shall be minimum 1-1/2 inches high and shall be of such size as to be legible. In addition, arrows and the word "UP", center of balance, sling or lifting point markings shall apply

5.1.2 Packaging of special tools Each set of special tools shall be boxed separately and shall not be included with the on board repair parts

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Purifiers covered by this specification are intended for use on board Naval ships for the removal of water and solids from fuel and lube oil

6.2 Acquisition requirements. Acquisition documents must specify the following

(a) Title, number, and date of this specification.
(b) Purifier size and operating fluid (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
(d) When first article inspection is required (see 3.1)
(e) When resilient mounts are required (see 3.3.2.1)
(f) Ship applicability (surface ship or submarine) (see 3.3.7).
(g) Piping connection requirements (see 3.3.7)
(h) When independent pump drive is required (see 3.3.20.1)
(i) When oil heaters are required, heating capability, and heater type (see 3.3.21 and 3.3.21.1)
(j) Motors (see 3.3.22)

(1) Voltage.
(2) Low noise application
(3) Type (ac, dc).

(k) Controllers (see 3. 3 23) {for surface ships only)

(1) Voltage
(2) Abnormal vibration frequency.
(3) Low noise application

(1) Shock requirements (see 3.4.4, 4.6.13, and 4. 6.13.2)
(m) Special tools (see 3.9)
(n) Noise requirements (see 4.6.12.2).
(0) Frequency range for vibration testing (see 4.6.11)
(p) Level of preservation and packing and packaging acquisition options (see 5.1)
(q) When fire-retardant packaging provisions apply (see 5.1(a))

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27 475-1 exempts the requirement for a DD Form 1423.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T08:08Z
Check the source to verify that this is the current version before use.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.6.1	DI-RELI-80250	Reliability test plan	-----
	DI-RELI-80251	Reliability test procedures	-----
3.6.1.2	DI-R-7085	Failure mode, effects, and criticality analysis report	-----
3.6 2	DI-MNTY-80831	Maintainability/test-ability demonstration test plan	-----
4.3.2 and 4.4	DI-T-23731	Notification of tests	-----
	DDI-T-23732	Procedures, test	-----
4 .4	DI-T-2072	Reports, test	
4.6.5	DI-RELI-80252	Reliability test reports	Task 202
4 6 6	DI-MNTY-80832	Maintainability/test-ability demonstration test report	Test method 4
Appendix A	DI-DRPR-80651	Engineering drawings	Level 2

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3.1 Bid data The following information should be submitted with each bid

(a) Manufacturer's model and type designation of purifier
(b) Sectional assembly drawing of purifier or reference to previously approved drawings.
(c) A drawing showing overall dimensions of the unit or reference to previously approved drawing
(d) Horsepower of driving motor.
(e) Output shaft speed of driving motor and of rotating element
(f) Weight (wet and dry) of complete unit.
(g) Manufacturers of major components such as motors and controllers
(h) Description of controls
(i) Prime water requirements.
(j) Solids retention volume
(k) List of special tools.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

6 6 Subcontracted material and parts The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.7.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.8 Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a related Government acquisition, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.9 Failure mode. effects and criticality analysis (FMECA). The analysis should include analyses of possible failure modes, their frequency, causes and effects, and precautionary measures to eliminate or minimize the failure modes. The depth of the FMECA should include all equipment, assemblies, subassemblies, and components whose failure would cause the purifier to fail to meet the requirements herein.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:08Z
Check the source to verify that this is the current version before use.

6.10 Reconditioning of shock-tested equipment Purifiers which have been subjected to high-impact shock tests and have successfully passed the test will be considered acceptable for such service as the contracting activity may authorize Such equipment will be returned to the contractor for reconditioning as follows

(a) Minor deformations of mounting feet corrected
(b) Minor deformations affecting alignment of parts or components corrected
(c) Bearings replaced
(d) Each part carefully examined by the contractor and any part which is considered substandard replaced.
(e) The rotating elements rebalanced.
(f) The motor corrected as required by the applicable motor specification.
(g) The reconditioned purifier- should be tested as specified in 4.6.2

Minor deformations are defined as those which do not. cause rejection of the equipment under high-impact shock test, but which are in excess of the dimensional tolerances specified on the applicable drawings

6.11 Special tools. One set of special tools as defined in 3 9 and 6 2 should be acquired for each ship’ s compartment in which purifiers are installed The number of sets required should be established by the acquisition activity

6.12 Subject term (key word) listing

Bearings
Brakes
Frames
Mounts
Ports, observation
Sight glass
Spindle

6.13 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Preparing activity
Navy - SH
(Project 4330-N055)

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:08Z
Check the source to verify that this is the current version before use.

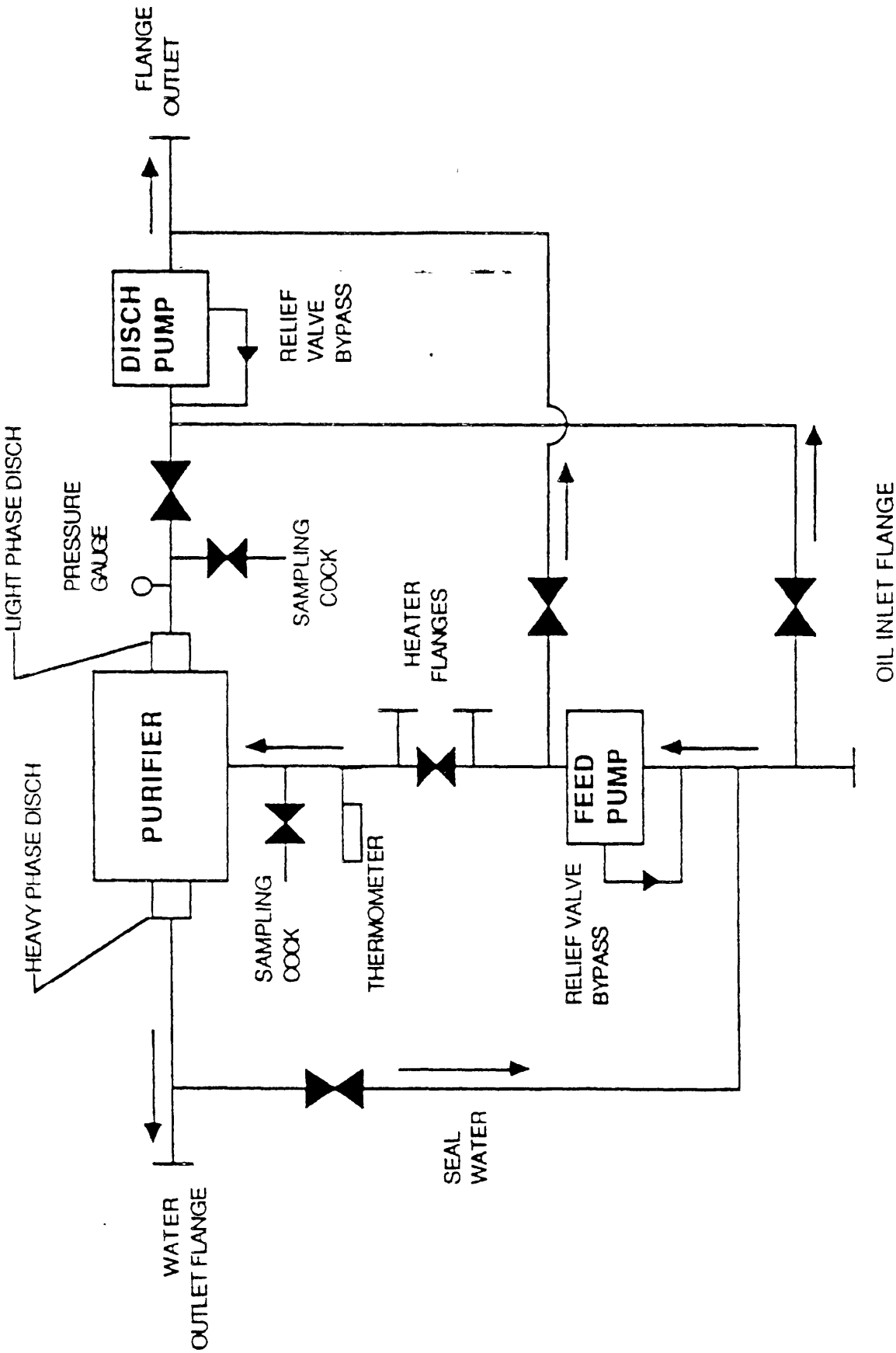


FIGURE 1 Diagram of purifier attached piping.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T08:08Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that should be included on the drawings when specified in the contract or order. This appendix is applicable only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications. standards. and handbooks The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless other-wise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION'S

FEDERAL

GGG-P-781 - Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. DRAWING CONTENTS

30.1 Drawings. The drawings shall contain the following characteristics:

(a) Sectional assembly drawing. Every part in the assembly shall be shown . Side or partial views may be used to show parts not otherwise shown in the main section view. All clearances, tightening torques, and thread lubrication requirements shall be shown on the assembly drawing.

(b) Complete parts list. Parts shall be named to indicate the function they serve. The parts list shall indicate each part recommended *to* be furnished as an on board repair part (this is not a repair parts list) and include the manufacturer's drawing number and service part number for all parts for which detail drawings are required. Standard threaded fasteners and washers, plugs, gaskets and o-rings shall be fully described as to specification, material, size, and thread fit (where applicable) . Such parts shall be identified by military standard part designation, where applicable

(c) Flow diagram and interface drawing showing power, pressure, temperature, and flow requirements (as applicable) for all connections to ship's systems.

(d) Controller drawings showing all indicators, alarms, and controls. It shall include a logic diagram and a parts list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

30.1.1 Outline drawings. Outline drawings and certification data shall be furnished as supplemental drawings to all sectional assembly and parts list drawings Separate outline drawings, drawing lists, and certification data shall be furnished under each contract unless the complete equipments covered by the outline drawings and the referenced drawings are identical in all respects Outline drawings, in addition to the requirements specified in the data ordering document, shall contain the following

(a) Dimensional outline assembly drawing of the purifier with its motor, bedplate, and attached auxiliaries.
(b) Table of weights of individual components and weight of complete unit (dry and in the operating, condition). This shall include the weight of the subbase, fluids, piping, and any other weight that may contribute to loading of the mountings
(c) Shipbuilder's connections showing size, type, and dimensions of flanges
(d) Radii of gyration of complete assembly about each of the three principal axes (required when equipment is sound isolated)
(e) Identification of h~gh-impact shock tests, nondestructive tests, and other tests, if performed, and the Command or agency letters approving these reports or extension of any tests
(f) The outline drawing shall include overall dimensions of the complete unit including locating dimensions of all fittings anti connections, the space required for removal and replacement of parts for maintenance, and mounting information
(g) Lifting instructions

30.1.2 "Manufacturer's Use Only" notes Information for manufacturer's use only shall be so designated

30.1.3 Preferred material reference. Where materials of identical or equal quality can be identified by more than one specification or standard, the drawings shall reference only one such specification or standard In selecting the specification or standard to be referenced, the following, is the order of preference

(a) Federal specification
(b) Military specification
(c) Industry and technical society specification or standard
(d) Manufacturer's specification or standard

30.1.3.1 Where materials other than those covered by (a) (b , and (c) of 30.1.3 are referenced, the drawings shall show the complete chemical and physical properties of the proposed material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military specification in order to enable Naval repair facilities to make emergency repairs as necessary from Navy material stocks.

Source: https://assist.dla.mil -- Downloaded: 20260-19T08:08Z
Check the source to verify that this is the current version before use.

30 1 4 Drawing identification Contractors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the part delineated thereon is produced by the subcontractor. The contractor shall not add his drawing number to the drawing, except as an unofficial reference outside the drawing border or margin

30 1 4.1 Contractors who purchase semi -finished parts from subcontractors for final production phases in their own shops have the option of using as the single drawing identification either their own title block and drawing number or the title block or drawing number of the subcontractor, but not both.

30.1.5 Other components. Drawings of drivers and other components where applicable shall be in accordance with the specifications covering those equipments.

30.1.6 Drawings of special tools Special tools shall be detailed on drawings and included in the list of materials Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identi fied as to type and use in notes on the tool drawings.

Source: https://assist.dla.mil -- Downloaded: 202640-19T08:08Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

- The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-20632B(SH)	20 APRIL 1992

3. DOCUMENT TITLE
PURIFIERS, CENTRIFUGAL, FUEL AND LUBE OIL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. Tom Marino (SEA 56Y32) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: 703-602-6830	(2) AUTOVON 332-6830
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026/10-19T08:08Z
Check the source to verify that this is the current version before use.